

Gaussian Elimination (partial pivoting)

Q1.

$$4x_1 + x_2 - x_3 = 2$$

$$5x_1 + x_2 + 2x_3 = 4$$

$$6x_1 + x_2 + x_3 = 6$$

$$\begin{bmatrix} 4 & 1 & -1 & | & 2 \\ 5 & 1 & 2 & | & 4 \\ 6 & 1 & 1 & | & 6 \end{bmatrix} \xrightarrow{\text{swapping}} \begin{bmatrix} 6 & 1 & 1 & | & 6 \\ 5 & 1 & 2 & | & 4 \\ 4 & 1 & -1 & | & -2 \end{bmatrix} \xleftarrow{R_2 - \frac{5}{6}R_1}$$

$$\xrightarrow{R_3 - \frac{4}{6}R_1} \begin{bmatrix} 6 & 1 & 1 & | & 6 \\ 0 & 1 - \frac{5}{6} & 2 - \frac{5}{6} & | & 4 - 5 \\ 0 & 1 & -1 & | & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 6 & 1 & 1 & | & 6 \\ 0 & \frac{1}{6} & \frac{7}{6} & | & -1 \\ 0 & 1 - \frac{2}{3} & -1 - \frac{2}{3} & | & -2 - 4 \end{bmatrix}$$

$$\begin{bmatrix} 6 & 1 & 1 & | & 6 \\ 0 & \frac{1}{6} & \frac{7}{6} & | & -1 \\ 0 & \frac{1}{3} & -\frac{5}{3} & | & -6 \end{bmatrix} \xrightarrow{\text{swapping}} \begin{bmatrix} 6 & 1 & 1 & | & 6 \\ 0 & \frac{1}{3} & -\frac{5}{3} & | & -6 \\ 0 & \frac{1}{6} & \frac{7}{6} & | & -1 \end{bmatrix} \xrightarrow{R_3 - \frac{1}{2}R_2}$$

$$\begin{bmatrix} 6 & 1 & 1 & | & 6 \\ 0 & \frac{1}{3} & -\frac{5}{3} & | & -6 \\ 0 & 0 & \frac{7}{6} + \frac{5}{6} & | & -1 + 3 \end{bmatrix} \rightarrow \begin{bmatrix} 6 & 1 & 1 & | & 6 \\ 0 & \frac{1}{3} & -\frac{5}{3} & | & -6 \\ 0 & 0 & 2 & | & 2 \end{bmatrix}$$

now,

$$2x_3 = 2 \quad \therefore x_3 = 1$$

$$\frac{1}{3}x_2 - \frac{5}{3}x_3 = -6$$

$$\Rightarrow \frac{1}{3}x_2 = -6 + \frac{5}{3} = \frac{-18+5}{3} \Rightarrow \frac{1}{3}x_2 = -13/3 \quad \therefore x_2 = -13$$

$$6x_1 + x_2 + x_3 = 6 \Rightarrow 6x_1 - 13 + 1 = 6 \quad \therefore x_1 = 3$$

$$\therefore \boxed{x_1 = 3, x_2 = -13, x_3 = 1} \quad (\text{Ans})$$

(Bisection Method)

02.

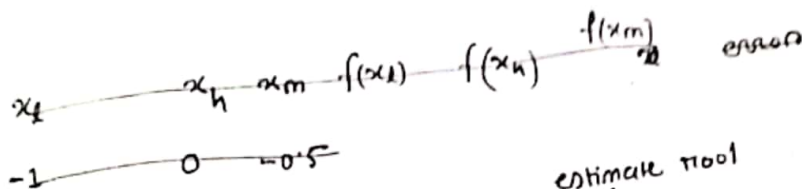
$$x^2 = 0.5 + \sin x$$

$$\therefore x^2 - \sin x - 0.5 = 0$$

initial guess = $[-1, 0]$

(x is considered as a radian angle)

$$\begin{aligned} &\{ (x_l) \} \{ (x_l) < 0, \quad x_l = x_l, \quad x_u = x_m \\ &\{ (x_u) \} \{ (x_l) > 0, \quad x_l = x_m, \quad x_u = x_u \end{aligned}$$



Iteration	x_l	x_u	estimate root (x_m)	$f(x_l)$	$f(x_m)$	relative error ↓ Ea %
01	-1	0	-0.5	1.39	0.23	100%
02	-0.5	0	-0.25	0.23	6.9×10^{-3}	33.33%
03	-0.5	-0.25	-0.375	0.23	-0.095	20%
04	-0.375	-0.25	-0.3125	6.9×10^{-3}	-0.045	9.09%
05	-0.375	-0.3125	-0.34375	6.9×10^{-3}		

03 (Newton-Raphson method)

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

$$\% = \left| \frac{x_{i+1} - x_i}{x_{i+1}} \right| \times 100$$

let $x = \sqrt[3]{10} \Rightarrow x^3 - 10 = 0 \therefore f'(x) = 3x^2$

iteration	x_i	$f(x_i)$	$f'(x_i)$	x_{i+1}	error %
01	3	17	27	2.37	26.56%
02	2.37	3.318	16.8507	2.17	9.06%
03	2.17	0.218	14.1267	2.15	0.7%

Am

04 ✓ (Newton's divided difference ^{method} for quadratic interpolation)

$$x = 12 + 3((98-1) \div 30)$$

$$= 12 + 3 \times 2 \sim 7$$

$$= 12 + 21$$

$$= 33$$

for $x=33$,

$$x_0 = 31, x_1 = 34, x_2 = 37$$

$$\therefore f_2(x) = b_0 + b_1(x-x_0) + b_2(x-x_0)(x-x_1)$$

$$b_0 = f(x_0) = f(31) = 55.67,$$

$$b_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \frac{f(34) - f(31)}{34 - 31} = \frac{33.92 - 55.67}{3} = -7.25$$

$$b_2 = \frac{\frac{f(x_2) - f(x_1)}{x_2 - x_1} - \frac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0} = \frac{\frac{92.81 - 33.92}{37 - 34} - (-7.25)}{37 - 31} = 4.48$$

$$\therefore f_2(33) = 55.67 - 7.25(33 - 31) + 4.48(33 - 31)(33 - 34) = 32.21$$

(Ans) ✓

Interpolation

Newton's divided difference

Lagrangian Interpolation

1. Newton's ~~div~~ divided difference interpolation doesn't require equal spacing.
2. It is faster to work out than Lagrang's.
3. Its logic is not as obvious and doesn't support inverse interpolation.
4. Newton's form is more efficient when one have to interpolate data incrementally.

5. Numerically stable. ✓

2. Lagrange's interpolation also doesn't require equal spacing
2. It is slower than Newton's divided difference interpolation.
3. It is easy to explain and works for inverse interpolation too.
4. Lagrange's form is more efficient when one have to interpolate several data sets on the same data points
5. Numerically unstable. ✓

6.

14, 68, 158, ...

consider it is a second order regression.

$$\hat{f}(x) = ax^2 + bx + c$$

 $x \quad y$
 $\sum x \quad \sum y \quad \sum x^2$

x	y	x^2	xy
1	14	1	14
2	68	4	136
3	158	9	474

$$\sum x = 6, \quad \sum y = 240, \quad \sum x^2 = 14, \quad \sum xy = 624$$

x	y	x^2	x^3	x^4	xy	x^2y
1	14	1	1	1	14	14
2	68	4	8	16	136	272
3	158	9	27	81	474	1422

$$\sum x = 6 \quad \sum y = 240 \quad \sum x^2 = 14 \quad \sum x^3 = 36 \quad \sum x^4 = 98 \quad \sum xy = 624 \quad \sum x^2y = 1708$$

$$\begin{bmatrix} 3 & 6 & 14 \\ 6 & 14 & 36 \\ 14 & 36 & 98 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 240 \\ 624 \\ 1708 \end{bmatrix}$$

$$\therefore c = -4, \quad b = 0, \quad a = 18$$

$$\therefore \hat{f}(x) = 18x^2 - 4$$

7.

$$I = \int_2^6 f(x)$$

$$n = 8$$

$$h = \frac{b-a}{n} = \frac{6-2}{8} = \underline{0.5}$$

$$f(x_0) = 5.04, \quad f(x_1) = 10.75, \quad f(x_2) = 18.97, \quad f(x_3) = 29.63,$$

$$f(x_4) = 42.16, \quad f(x_5) = 55.12, \quad f(x_6) = 65.63, \quad f(x_7) = 68.5$$

$$f(x_8) = 54.63$$

$$f(x_8)$$

$$\therefore \int_a^b f(x) dx = \frac{b-a}{3n} \left[f(x_0) + 4 \sum_{\substack{i=1 \\ i=\text{odd}}}^{n-1} f(x_i) + 2 \sum_{\substack{i=2 \\ i=\text{even}}}^{n-2} f(x_i) + f(x_n) \right]$$

$$= \frac{6-2}{3 \times 8} \left\{ 5.04 + 4 \times (10.75 + 29.63 + 55.12 + 68.5) + \right. \\ \left. 2 \times (18.97 + 42.16 + 65.63) + 54.63 \right\}$$

$$= \frac{1}{6} (5.04 + 656 + 253.52 + 54.63)$$

$$= 161.53$$

(Amy

3.

x	y
4	5800
5	5700
7	4200
8	4100
9	3100
10	2500

(~~linear~~ linear model)

બમલે લીન રાખે.
↓ એ ઊંચી મિલિટી લાગે (મિલિટી)

$$\left\{ \begin{array}{l} e^{ax} = 10968.51 e^{-0.139x} \\ a + \frac{b}{x} = 924.8 + \frac{21319.15}{x} \\ axb = 21852.3x \end{array} \right.$$

consider exponential model,

$$y = ae^{bx}$$

$$\ln y = a \ln a + bx$$

$$\downarrow$$

$$Y = a' + bx$$

$$b = \frac{n \sum x_i \ln y_i - \sum x_i \sum \ln y_i}{n \sum x_i^2 - (\sum x_i)^2}$$

$$a = \bar{Y} - b\bar{x}$$

x	y	ln	x	ln y	$x_i(\ln y_i)$	x_i^2
			4	8.666	34.66	16
			5	8.65	43.27	25
			7	8.343	58.4	49
			8	8.319	66.55	64
			9	8.07	72.35	81
			10	7.847	78.47	100

$$\sum n = 43 \quad \sum \ln y = 49.842 \quad \sum x_i(\ln y_i) = 353.47 \quad \sum x_i^2 = 335$$

$$b = \frac{6 \times 353.47 - 43 \times 49.842}{6 \times 335 - 43^2} = -0.14$$

$$a' = \frac{49.842}{6} + 0.14 \times \frac{43}{6} = 9.3$$

$$\therefore a = 11064.43$$

$$\therefore y = 11064.43 e^{-0.14x}$$

- 6300
- 5500
- 4150
- 3600
- 3138
- 2700